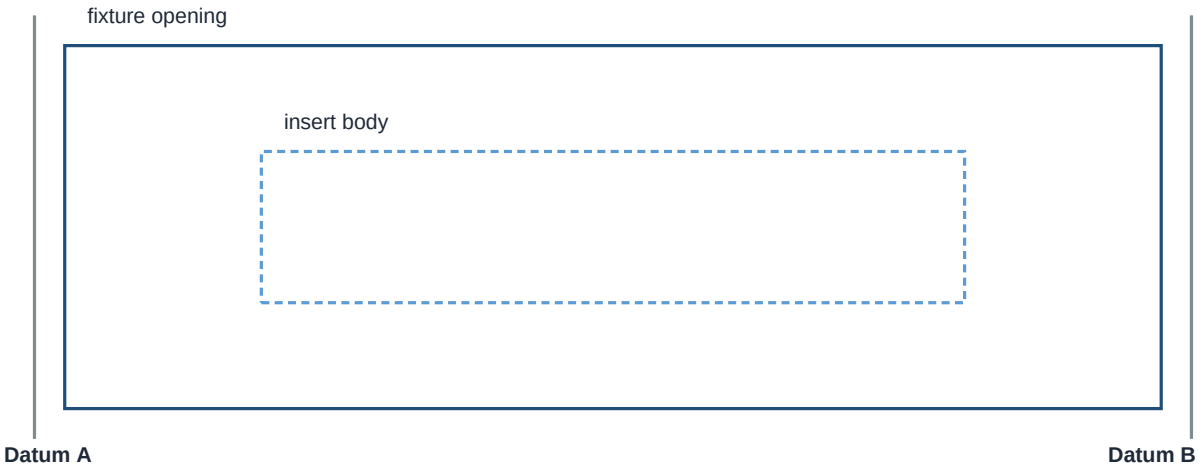
Cell retrofit clearance drawing

Drawing LCH-CELL-RETROFIT | Revision C | controlled 28 October 2025

PLAN VIEW - X and Y envelope



Dimension schedule

ID	View	Datum	Feature	Nominal	Minus	Plus	Unit	Related
D-101	PLAN	A|B	Fixture opening X	52.00	-0.20	+0.20	mm	D-201
D-102	PLAN	A|B	Insert body X	44.50	-0.10	+0.10	mm	D-202
D-103	PLAN	A|C	Fixture opening Y	30.00	-0.15	+0.15	mm	D-203
D-104	PLAN	A|C	Insert body Y	22.00	-0.05	+0.05	mm	D-204

Drawing notes

- All dimensions are millimetres. Tolerances are asymmetric where separately shown as minus and plus.
- Worst-case available clearance is derived from the lower opening bound minus the upper insert bound.
- The repeated related views on sheet 2 use the same datums and units as the plan callouts.
- Drawing reference for the controlling X envelope: D-101 and D-102; related view references D-201 and D-202.

Envelope calculation reference

X: (52.00 - 0.20) - (44.50 + 0.10) = 7.20 mm minimum available clearance

Y and Z remain visible on sheet 2; X is the controlling minimum envelope for the cell check.

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SECTION B / END C - related views and Z envelope

SECTION B

same A|B and A|D datums

END C

same A|C and A|D datums

Repeated callouts

ID	View	Datum	Feature	Nominal	Minus	Plus	Unit	Related
D-105	SECTION B	A|D	Fixture opening Z	18.00	-0.10	+0.10	mm	D-205
D-106	SECTION B	A|D	Insert body Z	10.00	-0.05	+0.05	mm	D-206
D-201	SECTION B	A|B	Fixture opening X repeat	52.00	-0.20	+0.20	mm	D-101
D-202	SECTION B	A|B	Insert body X repeat	44.50	-0.10	+0.10	mm	D-102
D-203	END C	A|C	Fixture opening Y repeat	30.00	-0.15	+0.15	mm	D-103
D-204	END C	A|C	Insert body Y repeat	22.00	-0.05	+0.05	mm	D-104
D-205	END C	A|D	Fixture opening Z repeat	18.00	-0.10	+0.10	mm	D-105
D-206	END C	A|D	Insert body Z repeat	10.00	-0.05	+0.05	mm	D-106

Derived envelopes

Direction	Opening lower bound	Insert upper bound	Available envelope
X / D-101 - D-102	51.80 mm	44.60 mm	7.20 mm
Y / D-103 - D-104	29.85 mm	22.05 mm	7.80 mm
Z / D-105 - D-106	17.90 mm	10.05 mm	7.85 mm

Control note

Related-view values are repeated callouts, not a second independent geometry. All repeated datums and tolerances agree.

The minimum physical envelope is 7.20 mm along X; no supplier datum is supplied on the separate acceptance snapshot.